

AI Floor Plan → 3D Virtual Tour

A web app that reads a 2D floor plan, lets you verify it, and builds an **empty 3D building** you can walk through in the browser — and later download as a video. This is the build order, tuned to run **free and local**.



Free & open-source



Runs locally · macOS



Empty building · no furniture



Apartments & villas



Solo build

01

Upload

Floor plan as JPG /
PNG / PDF

02

AI detects

Walls, rooms, doors,
windows

03

You verify

Fix & approve the 2D
plan

04

3D builds

Empty rooms, default
look

05

Tour & video

Web walkthrough + MP4

A The build roadmap

Build in this order. Each phase works on its own before you start the next — never build the 3D engine before the 2D plan is trustworthy.

P0

Setup & environment

~2–4 days

- Install the toolchain: Node, Python 3.11+, PostgreSQL, Redis, Blender, FFmpeg.
- Create the repo: a frontend (Next.js) and backend (FastAPI) folder.
- One docker-compose to start Postgres + Redis locally.
- Local file storage = a folder on disk (swap for MinIO later if you want S3-style URLs).

Node

Python

Docker

PostgreSQL

Redis

Blender

FFmpeg

DONE WHEN both apps start on localhost and talk to the database.

P1

Web app foundation

~2–3 weeks

- Email + password login, register, forgot-password, email verify.
- Dashboard: list of projects with status, date, preview, open/delete.
- Create-project wizard: project details → Apartment or Villa → building & room forms.
- Autosave + draft status. Store everything in Postgres.

Next.js

TypeScript

Tailwind

FastAPI

JWT auth

DONE WHEN a user can register and create a full draft project — no AI yet.

P2

Upload + AI floor-plan detection

~3–5 weeks

- Upload & validate plan files (type, size, resolution). Store + link to a floor.
- Run detection as a **background job** (Celery + Redis) so the web stays fast.
- Detect walls & rooms (OpenCV + segmentation), doors/windows (YOLO), labels & sizes (OCR).
- Output structured JSON: wall lines, room boxes, openings — **each with a confidence score**.

OpenCV

PyTorch

YOLO

Tesseract OCR

Celery

DONE WHEN a plan produces a JSON layout of walls, rooms, doors & windows.

P3

2D verification editor

~3–4 weeks

- Show the original plan with the AI vectors drawn on top (toggle opacity).
- Add / move / delete walls, doors, windows; rename & resize rooms.
- **Set the scale** (pixels → metres) — this makes dimensions real.
- Warnings panel blocks approval until fixed. Approve = locked source of truth.

React

Canvas / Konva

undo/redo

versioning

DONE WHEN the user can approve a clean, scaled 2D plan version.

P4

3D building generation (empty rooms)

~4–6 weeks

- Turn approved walls into 3D wall meshes; cut door & window openings cleanly.
- Add floors, ceilings, stairs, lifts, balconies, railings. **No furniture.**
- Apply the default theme: white walls, brown doors, glass windows, tiled floor.
- Stack multiple floors for apartments. Export a web-friendly model (glTF/GLB).

Blender Python API

procedural geometry

glTF export

DONE WHEN an approved plan produces a valid, hole-free 3D model file.

P5

Interactive web tour

~2–3 weeks

- Load the 3D model in the browser (Three.js / React Three Fiber) — **renders free, no GPU server**.
- Navigation: orbit, zoom, pan, walk / first-person, floor view, exterior.
- Auto walkthrough route through the rooms; show/hide floors; room info.
- Share by link + full-screen — this is the viewer style from your reference sites.

Three.js

React Three Fiber

WebGL

glTF loader

DONE WHEN anyone can open a link and walk the building in a browser.

P6

Video render

DO LAST

~2–3 weeks

- Move the camera along the tour route in Blender and render frames.
- Encode to MP4 (H.264) with FFmpeg — 1080p, in 16:9 and 9:16.
- Add logo, title, watermark. Render **locally on your Mac** (slow but free).
- This is the only heavy part — keep clips short & low-res while testing.

Blender render

FFmpeg

GPU worker

DONE WHEN a project produces a playable, downloadable MP4.

B Tech stack — all free

Every tool below is open-source and installs on your Mac. You pay ₹0 for software; you only "pay" in render time.

FRONTEND

Next.js + React	app & dashboard
TypeScript	safety
Tailwind CSS	styling
Three.js / R3F	3D viewer
Konva / Canvas	2D editor

BACKEND

Python + FastAPI	REST API
WebSocket	live status
Celery	background jobs
JWT	auth

AI / COMPUTER VISION

OpenCV	walls / rooms
PyTorch	models
YOLO	doors / windows
Tesseract OCR	labels / sizes

3D & VIDEO

Blender + API	build & render
glTF / GLB	web model
FFmpeg	MP4 encode

DATA

PostgreSQL	main data
Redis	cache / jobs
Disk / MinIO	files

INFRA (LOCAL)

Docker Compose	run it all
Git	version
Your Mac	the "server"

C Core data model

The key tables to create in Postgres. Approved plans and models are **versioned** — a change never overwrites, it makes a new version.

User

name · email · role

Project

name · type ·
location · status

Building

floors · height ·
entrances

Floor

level · height ·
plans

Room

type · w · l · h ·
connections

PlanFile

file · floor ·
version

DetectedElement

type · geometry ·
confidence

ApprovedPlan

vectors · scale ·
approver

Model3D

source plan · glb
file

RenderJob

status · output ·
usage

D Cut for now — build later

Keep the first version small. Skip these until the core loop works end-to-end.

⚠ Not in the first build

- **Furniture & interior design** — removed by your decision. Empty rooms only.
- **Video rendering (P6)** — do the free browser tour first; add MP4 last.
- **DWG / DXF / IFC / Revit** — start with JPG / PNG / PDF only.
- **Google / Microsoft SSO** — plain email + password is enough to start.
- **Payments, credits, multi-tenant admin** — not needed for a local build.
- **4K, voice-over, music, VR/AR headsets** — future phase.

E Start here — first 5 steps

Concrete actions for week one, before writing any AI code.

1

Install the tools

Node, Python 3.11+, Blender, FFmpeg, Docker Desktop. Confirm each with a `--version` check.

2

Scaffold the two apps

`npx create-next-app frontend` and a FastAPI `backend`. Get "hello world" on both localhost ports.

3

Start Postgres + Redis

A small `docker-compose.yml`. Connect FastAPI to the database and run your first migration.

4

Build login + project create

This is Phase 1 — a working app with no AI. It proves the whole skeleton before the hard parts.

5

Prototype AI on ONE plan

In a Python notebook, load one floor plan and detect walls with OpenCV. Prove the idea before wiring it in.

Reality check on "free & local": all software is free and runs on your Mac. The one slow part is video rendering (P6) and heavy AI — these work on your machine but take longer than a paid cloud GPU. Note: macOS uses Apple Silicon, not NVIDIA/CUDA — PyTorch (MPS) and Blender still run, but a few CUDA-only tools won't.

Disclaimer (from the BRD): the generated model and tour are visual representations only — not certified architectural, structural or legal documents. Dimensions and finishes may vary.

Rough solo timeline: ~4–6 months to a working MVP through Phase 5. Times assume one developer learning as they go — adjust to your pace.